

Class 10 NCERT Maths

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Chapter 6: Triangles (Core-Basics Edition)

CHAPTER 6: TRIANGLES

A simplified, detailed guide in Hinglish

Special edition for students who skipped Class 7-8-9

IS PDF MEIN RANG KA MATLAB:

RED text = core basic / prerequisite (Class 7-8-9 ka).

GREEN text = exam mein BAAR-BAAR aata hai (zaroor yaad rakh).

(Aur black = normal explanation.)

CORE BASICS - MISS MAT KARNA (RED PAGE)

1) TRIANGLE

3 side, 3 angle. Teeno angle ka sum = 180 degree.

2) RATIO AUR PROPORTION

Ratio = tulna ($4/6 = 2/3$).

Proportion = do ratio barabar ($a/b = c/d$).

3) CONGRUENT vs SIMILAR (bahut important fark)

Congruent = bilkul same (shape AND size).

Similar = same shape, size alag (sides PROPORTIONAL).

Photo aur uski zoom copy = similar.

4) CORRESPONDING (aamne-saamne wale)

Similar triangles mein corresponding angles equal aur corresponding sides proportional hote hain.

Order important: $ABC \sim PQR \rightarrow A=P, B=Q, C=R$.

5) PARALLEL LINES (||)

Do lines jo kabhi nahi milti. Symbol: ||.

6) SQUARE / SQUARE ROOT

$a^2 = a \times a$. $\sqrt{25} = 5$.

Pythagoras mein squares use hote hain.

7) PROPORTION SOLVE KARNA (cross multiply)

$a/b = c/d \rightarrow a \times d = b \times c$.

Isi se unknown side nikalti hai.

NOW: ASLI CHAPTER 6 SHURU

TOPIC 1: SIMILAR TRIANGLES

Do triangles similar hain agar:

1. Corresponding angles equal, AUR
2. Corresponding sides proportional.

Likha: triangle ABC ~ triangle PQR.

[CORE] "~" ka matlab similar. Order se corresponding parts pata chalte hain (RED page point 4).

TOPIC 2: BPT / THALES THEOREM

EXAM ALERT: BPT statement aur use HAR exam mein aata hai.

"Agar triangle ke ek side ke parallel line baaki do sides ko cut kare, toh wo dono sides SAME RATIO mein bant-te hain."

Agar $DE \parallel BC$ (triangle ABC mein), toh:

$$AD/DB = AE/EC$$

CONVERSE: agar $AD/DB = AE/EC$, toh $DE \parallel BC$.

TOPIC 3: SIMILARITY CRITERIA

EXAM ALERT: Ye criteria yaad rakh (similar prove karne):

AA : do angles equal -> similar (most used)

SSS : teeno sides proportional -> similar

SAS : do sides proportional + beech ka angle equal

TOPIC 4: PYTHAGORAS THEOREM

EXAM ALERT: Right triangle mein:

$$(\text{Hypotenuse})^2 = (\text{side1})^2 + (\text{side2})^2$$

CONVERSE: agar bada side² = baaki do ke squares ka sum, toh triangle right-angled hai.

[CORE] Triplets yaad rakh: (3,4,5), (5,12,13),

(8,15,17), (7,24,25). Calculation fast hoti hai.

SOLVED EXAMPLES: HARDEST -> EASIEST

8 examples poori tarah solve karke. Upar HARDEST, neeche EASIEST.

Solved Example 1 (HARDEST) - BPT do unknown

(Exam favourite)

Triangle ABC mein DE || BC. AD = x, DB = x-2, AE = x+2, EC = x-1. x nikaalo.

BPT: AD/DB = AE/EC

$$x/(x-2) = (x+2)/(x-1)$$

Cross multiply:

$$x(x-1) = (x-2)(x+2)$$

$$x^2 - x = x^2 - 4$$

$$-x = -4 \rightarrow x = 4.$$

Solved Example 2 - Similar triangles, side nikaalo

(Exam favourite)

triangle ABC ~ triangle DEF. AB=6, BC=8, CA=10, DE=9. EF aur FD nikaalo.

Sides proportional: AB/DE = BC/EF = CA/FD

$$6/9 = 8/EF \rightarrow EF = (8 \times 9)/6 = 12$$

$$6/9 = 10/FD \rightarrow FD = (10 \times 9)/6 = 15.$$

Solved Example 3 - Pythagoras (ladder/height)

(Exam favourite)

Ek ladder wall pe lagi hai. Foot wall se 9 m door, ladder ki length 41 m. Wall pe kitni unchai chhui?

$$h^2 + 9^2 = 41^2$$

$$h^2 = 1681 - 81 = 1600$$

$$h = 40 \text{ m.}$$

Solved Example 4 - SSS similarity check

triangle ABC: 4,6,8 ; triangle DEF: 6,9,12. Similar?

$4/6 = 2/3$, $6/9 = 2/3$, $8/12 = 2/3$ (sab same)
Haan, similar (by SSS).

Solved Example 5 - BPT simple

DE $\parallel$ BC. AD=2.5, DB=3, AE=4. EC nikaalo.
 $AD/DB = AE/EC \rightarrow 2.5/3 = 4/EC$
 $EC = (4 \times 3)/2.5 = 4.8$.

Solved Example 6 - Pythagoras hypotenuse

Right triangle, sides 6 aur 8. Hypotenuse?
 $h^2 = 6^2 + 8^2 = 36 + 64 = 100 \rightarrow h = 10$.

Solved Example 7 - Right angle check (converse)

Kya 5, 12, 13 right triangle banate hain?
Bada side 13: $13^2 = 169$.
 $5^2 + 12^2 = 25 + 144 = 169$ (same!)
Haan, right-angled triangle hai.

Solved Example 8 (EASIEST) - Angle sum

Triangle ke do angle 50 aur 60 hain. Teesra?
 $\text{Teesra} = 180 - (50 + 60) = 70$ degree.

Q AND A TIME

Q1. CORE: congruent aur similar mein fark ek line mein.

Q2. triangle PQR $\sim$ triangle XYZ. PQ=6, QR=8, RP=10,
XY=9. YZ aur ZX nikaal.

Q3. DE $\parallel$ BC. AD=1.5, DB=3, AE=1. EC nikaal (BPT).

Q4. Right triangle ke sides 9 aur 12. Hypotenuse nikaal.

Q5. Kya 8, 15, 17 right triangle ke sides hain? Check kar.

Q6. Konsa similarity criterion: 2 angles equal hain?

Q7. CORE CHECK: triangle ke do angle 90 aur 45 hain.
Teesra angle?

SUMMARY

1. Similar = same shape, sides proportional, angles equal.
2. BPT: $DE \parallel BC \rightarrow AD/DB = AE/EC$.
3. Criteria: AA, SSS, SAS.
4. Pythagoras: $\text{hyp}^2 = \text{side1}^2 + \text{side2}^2$.
5. Triplets: (3,4,5), (5,12,13), (8,15,17), (7,24,25).

CORE (RED page) revise: ratio/proportion, congruent vs similar, corresponding parts, cross multiply, squares.